

The Accountability Gap in Autonomous Defense Systems: An Ethical Audit of U.S. Regulations and Engineering

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Autonomous and algorithmically governed defense systems are being deployed faster than the regulatory and professional frameworks meant to govern them. This produces what the commentary refers to as the accountability gap, the legal and ethical space in which harm caused by an autonomous system cannot be traced to a single human decision. Drawing from an audit of U.S. defense policies and two real life case studies, this commentary argues that current regulation is organized around procedural oversight rather than substantive ethical accountability. We analyze both cases through utilitarian, deontological and care ethics frameworks to show how ethics can still be violated with procedural compliance. We then propose four targeted reforms based on our analysis of audits and identification of ethical gaps. The central contribution of this commentary is to relocate ethical responsibility upstream, into the engineering choices that determine whether a system can fail ethically in the first place.

Keywords – *autonomous weapons; engineering ethics; accountability gap; meaningful human control; defense regulation; lethal autonomous weapon systems*

1 Introduction

The defense industry is developing autonomous weapons at a faster pace than the regulatory and ethical frameworks designed to govern them. These contemporary systems rely on machine learning models that update continuously after deployment, meaning an algorithm can refine how it identifies targets during an active mission. However, existing defense ethics frameworks were written for an era of human-directed weapons and have not kept pace with the monumental growth of artificial intelligence, autonomous targeting and adaptive systems in the battlefield. This erodes a line that engineering ethics has long depended on, which is the separation between a system under test and a system in operational use. When a weapon learns in the field, the battlefield functions as the laboratory, and the subjects of that experiment are people who have not consented to participate.

This status quo produces a misalignment that falls directly onto an individual engineer, whose professional code is also not designed to resolve such issues. Frameworks like the National Society of Professional Engineers' (NSPE) Code of Ethics are grounded in the duty to protect public welfare (NSPE, 2019). This duty assumes there is a public that the engineer can clearly identify, measure harm against and be held accountable to, but these assumptions are unsettled by autonomous defense systems. The consequences of a deployed system are difficult to quantify when the system learns on the go after release; the engineer's specific duties become unclear when lethal decisions are delegated to an algorithm; and the population most affected by the design lives outside the jurisdiction in which the system is being built. This means that they hold no legal standing or political

representation in the geographical area where the engineers design and deploy such systems. The appeal to national security is often invoked as justification for these actions, but it offers little practical guidance on where the engineer's duty begins or ends.

From this misalignment emerges what the commentary will refer to as the *accountability gap*, which is the legal and ethical space in which harm caused by an autonomous system cannot be traced to a single human decision (Adamson, 2026). When a loitering munition misidentifies a civilian vehicle and engages it, responsibility ranges from the engineer who trained the targeting model to the commander who authorized the mission and the procurement process that certified the system. No single party can be attributed for the outcome, and hence this points to a deeper absence of ethical frameworks built for engineering itself.

This commentary will set out to audit the current US and international regulatory landscape governing autonomous defense systems. It will analyze two documented incidents where existing law and professional ethics fall short and identify the ethical gap. From this analysis, concrete reforms are proposed to both deferral policy and engineering codes, with the aim of restoring a clearer line of responsibility between the engineer and the outcomes their work produces.

2 Regulatory and Policy Audit

Current international regulations and policies are not very well developed. The UN Office for Disarmament Affairs discusses Lethal Autonomous Weapons (LAWs) in their Convention on Certain Conventional Weapons (CCW) with a 2025 GGE session finding difficulty with establishing consensus among many nations' implementation of LAW regulation. There were significant disagreements on the definition and usage of various keywords and qualification of "context-appropriate human control and judgement", which was contested and obscured by various undefined direct and indirect forms of human control (notably absent is the discussion of the effect of artificial intelligence on human intervention and usage). As such, our policy audit is based on the guidelines established by US' Department of Defense, being one of the premiere figures on LAWs and due to the lack of widely accepted international regulatory codes or even a consensus on definition of lethal autonomous weaponry.

The Department of Defense outlines its policy on autonomous weapons systems under DoD Directive 3000.09, which aims to establish policy, responsibilities, and guidelines for the development and usage of autonomous weaponry. This audit is respective to the effective January 23, 2023, release. In section 1.2, they outline their policy on the usage of Autonomous weapons, described below and detailed in section 3 as follows:

"Autonomous and semi-autonomous weapon systems will be designed to allow commanders and operators to exercise appropriate levels of human judgment over the use of force."

Critically, there is no clear definition of what constitutes 'appropriate levels of human judgement.' Evaluation of autonomous systems primarily focuses on the reliability of AI targeting, but details on the level of human judgement required are sparse, with responsibility relegated to individual military training programs and undefined doctrines. Their definition of an autonomous weapon system is one which automates both selection and engagement. This includes but is crucially not limited to operator-supervised, overridable systems, of which, the degree of supervision is unclear per the vague definition of appropriate judgement and could extend so far as allow loitering munitions to operate without significant supervision so long as the loitered area is judged to be appropriately operable.

Two regulatory instruments address whether autonomous systems can be tamper-resistant and correctly engage their intended targets. The first is DoDI 8500.01, applied through Section 4's review process. The second is the 2026 National Defense Authorization Act (NDAA), which introduces several provisions governing adjacent AI technologies used in autonomous weapons. Section 6602 requires each intelligence community to track the "efficacy, safety, fairness, transparency, accountability, appropriateness, lawfulness, and trustworthiness" of these

systems. Sections 6603 and 1533 establish testing and security standards for AI models, framed under "ethical principles regarding the use of artificial intelligence models" within the intelligence communities and the DoD.

In regard to legislative and regulatory oversight, section 4 of the directive establishes guidelines for senior review and approval and development by the Under Secretary of Defense for Policy, Under Secretary of Defense for Research and engineering, and the Vice Chairmen of the Joint Chiefs of Staff, however a waiver can be requested to the Deputy Secretary of Defense under exigent circumstances. Congressional oversight of all waivers issued within 30 days is established in section 1061 of the 2026 NDAA; however, this oversight is retrospective. In addition, such a review by governing bodies does not apply to the exceptions provided under section 1.2.d of the directive which outlines several types of weapon systems which can automatically proceed with formal development and fielding. Most notable among these is 1.2.d.1, which involves systems classified as a "semi-autonomous weapon." The directive broadly defines semi-autonomous weapons as systems where the act of engagement is done by an operator but can otherwise be autonomous in all other functions including the determination of target groups, to quote: "identifying potential targets" and "cuing potential targets." This is particularly concerning because it calls into question the cogency of the operator's role to engage when the operator's decision is enacted upon variables constructed by the system.

This audit finds a distinct lack of attention in establishing guidelines for and considering the level of autonomy acceptable for deploying autonomous systems, instead primarily implementing formal development and fielding procedures. This presents a major ethical gap in the regulation of autonomous weaponry, a state of affairs that appears to be widely present in the larger global sphere, already manifested in the case studies shown below.

3 Ethical Case Studies and Ethical Analysis

3.1 2021 Kabul Drone Strike

On 29 August 2021, during the final days of the US withdrawal from Afghanistan, a U.S. MQ-9 Reaper drone operated by U.S. Central Command's Over-the-Horizon Strike Cell launched a Hellfire missile at a white Toyota Corolla in a residential Kabul neighborhood. The strike killed ten civilians, including seven children. The driver, Zemari Ahmadi, was an electrical engineer employed by Nutrition and Education International, a California-based humanitarian NGO (Speri, 2023) and not an ISIS-K operative as initially claimed.

The targeting chain illustrates a cascading failure of interpretation along the chain of command. U.S. intelligence officials had tracked Ahmadi's movements for hours, flagging his "erratic route", and concluded that his vehicle contained explosives (Hennigan, 2021). In reality, Ahmadi and a co-worker had been filling empty plastic containers with water to bring home during a neighborhood water shortage. What analysts believed was an ISIS-K operative was a longtime aid worker; what they believed was a terrorist safe house was a home full of children; and what they believed were explosives were water bottles. Crucially, human operators had supervised every stage of this kill command, yet confirmation bias transformed what was ambiguous surveillance data into lethal certainty.

The aftermath exposes the accountability gap concretely. The Department of Defense initially insisted the strike was legitimate, citing fabricated "secondary explosions", and only retracted its position after a New York Times visual investigation disproved the official account (Koettl et al., 2021). General Kenneth McKenzie publicly accepted responsibility, yet no U.S. personnel faced disciplinary action, prompting fifty legislators to call for a review of military practices. Promised condolence payments to surviving relatives remain undelivered nearly two years after the incident.

Kabul represents the ceiling, not the floor, of human-in-the-loop safeguards. If a fully supervised kill chain with hours of deliberation produced this outcome, scaling a similar architecture to algorithmic speeds, as seen in the next case study, compounds rather than mitigates these failure modes.

3.2 Lavender AI Targeting System

During the opening weeks of the Gaza war, the Israel Defense forces deployed an AI-based targeting program called “Lavender” to mark Palestinians suspected of Hamas or Palestinian Islamic Jihad affiliation (Abraham, 2024). At one point, the system listed as many as 37,000 Palestinian men as potential targets. In contrast to prior doctrine where lethal strikes were reserved for senior officials, Lavender extended target selection to low-ranking operatives, dramatically scaling up the kill chain.

The underlying engineering exposes the ethical failure. Lavender employs a form of semi-supervised machine learning called “positive unlabeled learning”, which trains an algorithm from a dataset containing labelled positives and unlabeled data to identify patterns across a larger population (Human Rights Watch, 2024). The system ranks individuals by features such as communication patterns and social proximity to known militants, with an adjustable threshold for what constitutes a probable target. Officers reportedly treated Lavender’s outputs as a formality rather than a genuine assessment of risk, with sweeping approval granted to its kill lists within examination of the underlying intelligence. Officers admit devoting roughly twenty seconds to authorize individual strikes, despite knowing that Lavender misidentifies targets in approximately 10 percent of all cases (Reiff, 2024). A companion system would then track flagged individuals until they returned home, where strikes were executed outside of military battlefields. These operations were carried out often with unguided “dumb” bombs that maximized collateral damage to entire families.

This case poses a striking difference to the Kabul failure mode. Whereas Kabul demonstrated that fully supervised human judgement can still fail, Lavender demonstrated that formal human oversight can exist but still be operationally hollow. The twenty second authorization window is not meaningful human control, but instead human control reduced to just ceremony. The engineering choices, from probabilistic scoring to an adjustable acceptance threshold, residential targeting logic and the absence of an audit trail requirement show that the system architecture is encoded for a tolerance to civilian death. The accountability gap here is not the product of rushed judgement under pressure as in Kabul, but of deliberate design decisions that diffuse moral responsibility across the black box of algorithmic approvals.

3.3 Ethical Analysis

The two cases are evaluated below against three ethical frameworks to expose how each regulatory safeguard fails in practice:

Case Study	Utilitarianism	Deontology	Care Ethics
3.1	The Over-the-Horizon cell traded force-protection gains for degraded intelligence quality, producing a catastrophic net harm outcome. Zero combatants were neutralized, ten civilians were killed and strategic damage was dealt to the credibility of U.S. counterterrorism.	Analysts violated their duty of distinction by inferring culpability from metadata, classifying routine errands as hostile intent. The targeting chain treated Ahmadi as a means to avert a hypothetical attack, not as a person who should have been verified.	The Over-the-Horizon architecture is designed to sever the operator from the human context of the strike. It increases the geographical distance between the operator and victims, ignores context in favor of efficiency and reduces any empathy in action.
3.2	Lavender’s scale and error rate amplify human suffering rather than reduce it. This optimizes the authorization of harm, while the use of “dumb” bombs compounds collateral damage beyond any proportionate military gain.	The probabilistic targeting of suspected low-ranking operatives violates IHL’s proportionality principle. This principle prohibits collateral damage exceeding anticipated military advantage, which is	Lavender reduces humans to data points and ignores context in favor of efficiency. Targeting individuals inside private homes violates the interpersonal obligations owed to vulnerable civilian populations

		impossible given Lavender’s 10% misidentification rate.	surrounding suspected combatants.
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4 Ethical Gaps and Solutions

The preceding audit and case studies expose a recurring pattern. Regulation in the United States is organized around procedural oversight of autonomous systems, like a list of checkboxes to be ticked off at each stage. However, each safeguard examined above failed precisely at the point where ethical accountability was most needed. This section identifies four ethical gaps that emerge from this pattern and proposes targeted reforms to both federal policy and the engineering profession's own codes of practice.

4.1 Undefined Threshold of “Appropriate” Human Judgement

DOD Directive 3000.09 requires operators to exercise “appropriate levels of human judgement over the use of force” but offers no operational definition of how that judgement should look like. The Kabul strike demonstrates that hours of supervised deliberation can still produce a catastrophic misidentification, while Lavender demonstrates that formal oversight can be compressed into a twenty-second approval window and still satisfy the procedural formalities. There is no clear line on what constitutes appropriate human judgement in different use cases and what meaningful judgement entails.

Recommended Action: It is proposed that the directive be amended to define minimum standards for human review, including minimum time threshold for target verification, independent third-party corroboration of intelligence inputs, and explicit documentation of the evidentiary basis for each strike. Such standards ensure that human judgement functions as a safeguard, rather than just a signature on a piece of paper.

4.2 Absence of Engineer-Level Accountability in Procurement and Certification of LAWs

Section 6602 of the 2026 NDAA requires intelligence communities to track the “efficacy, safety, fairness, transparency, accountability, appropriateness, lawfulness, and trustworthiness” of AI systems. However, these requirements fall on the agency that deploys and uses them, not on engineers who design and validate the system. Lavender’s 10% misidentification rate was known to its designers before deployment, yet no regulatory mechanism required them to publicly justify that error rate despite the fact that it could amplify largely across tens of thousands of real human targets.

Recommended Action: We propose that professional engineering codes, such as the NSPE Code of Ethics, be extended to include a duty of *foreseeable harm disclosure* for engineers working on autonomous lethal systems. This duty would require engineers to document and disclose certain information to an independent review body. This would include information like the expected error rates of their designs, the populations most likely to be affected by those errors, and the engineering choices that determine the system’s tolerance to civilian harm. This mirrors existing disclosure obligations in a field like civil engineering, where structural tolerances and failure models must be declared before a design can be certified.

4.3 The Semi-Autonomous Loophole

Section 1.2.d.1 of DoDD 3000.09 exempts semi-autonomous weapons from the senior review process that is required for autonomous systems. It however defines “semi-autonomous” broadly enough to include systems that independently identify and cue targets so long as a human performs the final act of engagement, just like the Lavender AI previously discussed. The exemption assumes that the human act of engagement itself it a substantive safeguard, but both case studies show that when the decision is enacted upon variables constructed by the system, the operator’s role becomes one of just procedural rectification.

Recommended Action: We propose that the semi-autonomous exemption be narrowed such that systems which perform target identification, selection or cueing algorithmically are excluded. Such systems should be subject to the same senior review process as fully autonomous weapons, on the principle that regulatory oversight should follow where the decision is made, not where the physical trigger is pulled.

4.4 Lack of Post-Incident Accountability Mechanisms

The Kabul strike produced no disciplinary action despite a public acknowledgement of error, and promised condolence payments remain undelivered. Lavender's operational record has never been subject to external audit. Current regulation contains no mechanisms by which affected civilian population or independent investigators can demand for disclosure of engineering decisions that produce a given harm.

Recommended Action: We propose the establishment of an independent Autonomous Systems Accountability Board, which will be analogous in function to the National Transportation Safety Board. It will have authority to investigate incidents involving autonomous or algorithmically mediated lethal force, subpoena relevant engineering documentation and publish its own independent findings. Engineers working on these systems would be required by duty to cooperate with investigations as a condition of professional licensure.

Taken together, these suggestions share a single principle, that accountability for autonomous systems must be distributed at the design stage rather than reconstructed after a harm has already occurred. The current regulations treat ethical failures as downstream problems to be managed through post-hoc review, waivers and condolence payments. These case studies show that this approach has reached its limit. Closing the accountability gap requires moving ethical scrutiny upstream, into design and engineering choices which determine whether a system can fail ethically in the first place.

5 Conclusion

Autonomous defense systems have outpaced the regulatory and professional structures meant to govern them. The audit of U.S. and global policy in Section 2 shows that current regulation is built around procedural oversight, with key terms such as "appropriate human judgement" left undefined and entire categories of semi-autonomous systems exempted from senior review. The case studies in Section 3 show what this regulatory environment produces: a fully supervised strike in Kabul that killed ten innocent civilians without disciplinary consequence, and an algorithmic targeting system in Gaza that compressed human oversight into a twenty-second formality despite being known for a high misidentification rate. In both cases, procedures were followed yet ethics still failed.

The accountability gap exposed by these incidents is not a problem that can be solved alone by legislative reform. It is fundamentally, a gap in how the engineering profession defines its own duties. Engineers who design these autonomous lethal systems are expected to satisfy procurement specifications and internal safety reviews. They are not expected to disclose foreseeable harm, justify error rates against human lives, or accept post-incident scrutiny of their design decisions. The reforms proposed in Section 4 are an attempt to relocate ethical responsibility upstream, into the engineering choices that determine in the first place, if a system can fail ethically.

Closing this accountability gap will require the engineering profession to recognize that in this domain, neutrality is not a choice. The decisions that engineers make at the design stage, are the decisions that determine who lives.

6 References

Abraham, Y. (2024, April 3). "Lavender": the AI Machine Directing Israel's Bombing Spree in Gaza. +972 Magazine. <https://www.972mag.com/lavender-ai-israeli-army-gaza/>

Adamson, L. (2026, March 9). Legal Accountability for AI-Driven Autonomous Weapons. Lieber Institute West Point. <https://lieber.westpoint.edu/legal-accountability-ai-driven-autonomous-weapons/>

Hennigan, W. J. (2021, September 17). “A Tragic Mistake.” Botched Drone Strike in Afghanistan Raises Concerns Over Biden’s Counterterrorism Strategy. Time. <https://time.com/6099377/afghanistan-drone-strike-counterterrorism/>

Koettl, C., Hill, E., Aikins, M., Schmitt, E., Tiefenthäler, A., & Jordan, D. (2021, September 10). Video: “Imminent Threat” or Aid Worker: Did a U.S. Drone Strike in Afghanistan Kill the Wrong Person?. The New York Times. <https://www.nytimes.com/video/world/asia/100000007963596/us-drone-attack-kabul-investigation.html>

NSPE. (2019). NSPE Code of Ethics for Engineers | National Society of Professional Engineers. Nspe.org; National Society of Professional Engineers. <https://www.nspe.org/career-growth/nspe-code-ethics-engineers>

Questions and Answers: Israeli Military’s Use of Digital Tools in Gaza. (2024, September 10). Human Rights Watch. <https://www.hrw.org/news/2024/09/10/questions-and-answers-israeli-militarys-use-of-digital-tools-in-gaza>

Reiff, B. (2024, April 25). Why human agency is still central to Israel’s AI-powered warfare. +972 Magazine. <https://www.972mag.com/israel-gaza-lavender-ai-human-agency/>

Speri, A. (2023, May 17). The U.S. Still Owes Money to Family of 10 Afghans It Killed in “Horrible Mistake.” The Intercept. <https://theintercept.com/2023/05/17/kabul-drone-strike-survivor-payment/>